

Enes Altun

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CONTACT

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SKILLS

- AI & ML**
 - ML/LLM deployment on local systems
 - Multimedia: Fast multimodal data processings via GStreamer
 - Parallel data processing: Apache Arrow, DataFusion
- Graphics**
 - Graphics programming: wgpu, wgsl/glsl
- Domain Expertise**
 - Cognitive Science, ML, EEG signal processing, Psychophysics

🎓 EDUCATION

- 2020 ● **Akdeniz University, Department of Psychology**
B.A. in Psychology 📍 Antalya, Turkey
- 2022 ● **Akdeniz University, Department of Psychology**
M.A. in Psychology 📍 Antalya, Turkey
 - **Thesis:** *Visual processing of different-race and great-ape faces in the context of spatial frequency: an ERP study.* [Visual Cognition](#) (2024)

🖥️ RESEARCH EXPERIENCE

- 2017 — 2021 ● **Undergraduate Research Assistant**
Akdeniz University Cognitive Neuroscience Lab 📍 Antalya, Turkey
 - Designed and conducted psychophysics experiments using EEG and signal processing.
 - Built small electronic devices (GSR sensors, robotic fingers) for psychophysical experiments.
- 2021 — 2023 ● **Graduate Research Assistant**
Akdeniz University Cognitive Neuroscience Lab 📍 Antalya, Turkey
 - Image and signal processing, human and computer vision, machine learning.

⚡ INDUSTRY EXPERIENCE

- Jul 2023 — Present ● **Rust Developer**
[Empatik AI](#) 📍 Stockholm, Sweden
 - Built a cross platform, GPU accelerated multimodal desktop app in Tauri.
 - Worked on high perf multimedia and on-device AI in Rust + ONNX Runtime for efficient local inference.
 - Optimized GPU & CPU utilisation across multiple ML/LLM models to minimise latency and memory overhead.
 - Developed real time parallel audio/video pipelines with GStreamer that integrate with AI models.

Altun, E., Er, M. N., & Gülbetekin, E. (2024). The early visual processing of faces in a basic classification task: an ERP study of spatial frequency, other-race and other-species effect. *Visual Cognition*, 1–24. doi.org/10.1080/13506285.2024.2415721

Gülbetekin, E., **Altun, E.**, Er, M. N., Fidancı, A., Keskin, P., & Steenken, D. (2021). Effects of right or left face stimulation on self and other perception in enfacement illusion. *Social Neuroscience*, 16(2), 189–205. doi.org/10.1080/17470919.2021.1886983

Gülbetekin, E., Bayraktar, S., Kantar Gül, D., Varlık Özsoy, E., Er, M. N., **Altun, E.**, & Fidancı, A. (2023). Does tactile stimulation of the face affect the processing of other faces? Neural and behavioural effects of facial touch. *Social Neuroscience*. doi.org/10.1080/17470919.2023.2245126

Gülbetekin, E., Fidancı, A., **Altun, E.**, Er, M. N., & Gürcan, E. (2023). Effects of mask use and other-race on face perception, emotion recognition, and social distancing during the COVID-19 pandemic. *Asian Journal of Social Psychology*, 1–16. doi.org/10.1111/ajsp.12570

Gülbetekin, E., Fidancı, A., **Altun, E.** (2023). How do we process faces from different ethnic groups and species? *Society for Neuroscience Annual Meeting*, Washington, DC.

Gülbetekin, E., **Altun, E.**, Er, M. N. (2020). Does own-species matter? Holistic face perception for great apes. *International Behavioral Neuroscience Society (IBNS) Online Poster Sessions*, Texas, USA, 3–7 August 2020, pp. 1–1.

Altun, E., Er, N., Gülbetekin, E. (2019). Does genetic similarity matter? Face-recognition performance for primates. *Fechner Day 2019: 35th Annual Meeting of the International Society for Psychophysics*, Antalya, Turkey, pp. 91–92.

Er, N., **Altun, E.**, Fidancı, A., et al. (2019). Effect of stimulated face side on multisensory perception. *Fechner Day 2019*, Antalya, Turkey, pp. 79–79.

Altun, E. (2023). altunenes/scramblery: 2.1.1.1. *Zenodo*. doi.org/10.5281/zenodo.7484576

Altun, E., Er, M. N., Gülbetekin, E. (*Under review*). Exploring the distinctiveness of early visual processing in human and illusory faces: an ERP study of spatial frequency effects. SSRN: ssrn.com/abstract=4341900

SELECTED OPEN-SOURCE PROJECTS

2024	Cuneus GPU compute engine for creating and shipping WGSL shader programs. Built with wgpu and GStreamer for real-time video/audio processing; multi pass pipelines and hot shader reload.  Rust · WGSL
2025	parakeet-rs Very fast ASR library wrapping NVIDIA's Parakeet models via ONNX Runtime. Real time streaming and speaker diarization.  Rust
2022	Scramblery Cross platform GPU accelerated desktop app for advanced image and video scrambling (Fourier phase, diffeomorphic warping with optical flow). Facial detection and efficient frame processing via GStreamer.  Rust
2023	Butter2D Pure Rust implementation of the Butterworth filter for spatial frequency filtering of images, with an online WASM demo.  Rust
2023	Weber–Fechner Template for psychophysics experiments implementing the Weber–Fechner law of perception, using the Bevy engine with WASM support.  Rust